



Welcome and tutorial overview

ISMB-ECCB 2025: Tutorial IP2 (MPRAs and IGVF Resource)

Martin Kircher

Regulatory Genomics @ UzL/UKSH, Computational Genome Biology @ BIH

July 20, 2025



UNIVERSITÄT ZU LÜBECK



UNIVERSITÄTSKLINIKUM
Schleswig-Holstein

BIH Berlin Institute
of Health
@Charité

Welcome to Liverpool and the ISMB-ECCB 2025



33rd CONFERENCE ON

**Intelligent Systems
for Molecular
Biology**

DATES: July 20 – 24

AND THE 24th

**European Conference
on Computational
Biology**

Tutorial IP2: Massively parallel reporter assays in functional regulatory genomics and as part of the IGVF data resource

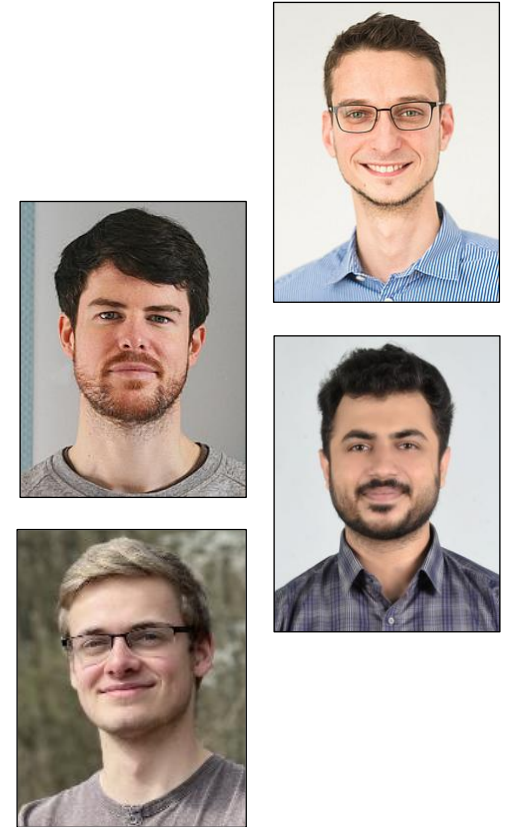
Start: **09:00**, End: **17:30**, Intended Audience and Level: **Beginner**

Speakers:

- **Martin Kircher, University of Lübeck / UKSH & Berlin Institute of Health at Charité**
- **Max Schubach, Berlin Institute of Health at Charité**
- **Arjun Devadas, University of Lübeck / UKSH**
- **Kilian Salomon, Berlin Institute of Health at Charité**

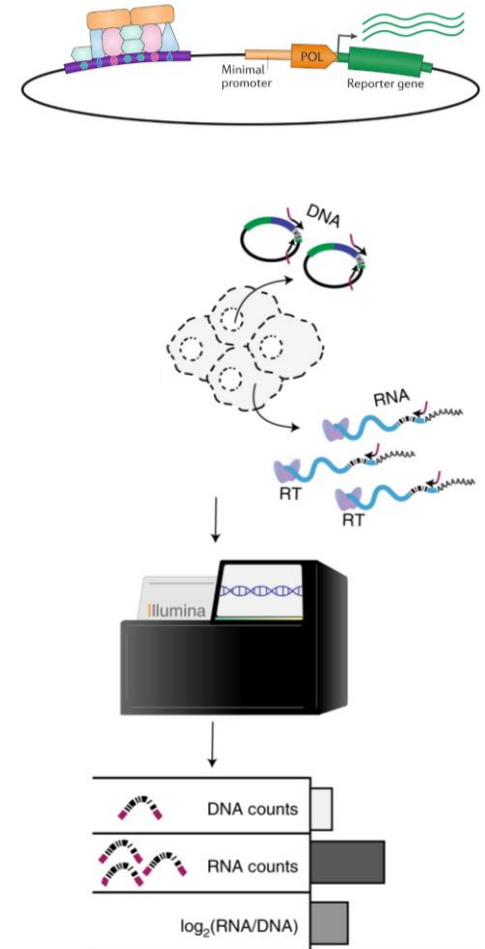
Support in preparation from:

- **Michael Love, University of North Carolina**
- **Jonathan Rosen, University of North Carolina**
- **Pia Keukeleire, University of Lübeck / UKSH**
- **Pyaree-Mohan Dash, Berlin Institute of Health at Charité**
- **Thorben Maaß, University of Lübeck / UKSH**



Learning Objectives

- Introduction to IGVF and how to access its data.
- MPRA Experiments: Explanation of experimental design, including the association of barcode/tag sequences to tested regulatory sequences and variants.
- Step-by-step data processing using MPRAsnakeflow, including construction of count tables and interpretation of quality control metrics.
- Statistical analysis of DNA and RNA barcode counts using BCalm.
- Applications of MPRA data like identifying active regulatory regions and significant variant effects, optimising cell-type-specific expression through synthetic sequences.
- Sequence-based modeling of regulatory sequence activity, i.e. using deep learning models to predict open chromatin and MPRA activity, investigate transcription factor motif importance and composition.



Schedule overview (1)

09:00-09:15	Welcome and introduction to the schedule
09:15-09:30	Tutorial materials access and setup of the Google CoLab environment Participants that fail with setting up their access und CoLab will be brought outside of the seminar room and helped by other speakers while the theoretical introductions commence
09:30-09:50	Lecture: IGVF and its data access Introduction to the IGVF mission and how to access its data
09:50-10:15	Lecture: Fundamentals of MPRA Experiments Explanation of experimental design, including the association of barcode/tag sequences to tested regulatory sequences and variants.
10:15-10:45	Coffee break

Schedule overview (2)

10:45-11:05	Hands-on: Association of Barcode/Tag Sequences Participants will learn how to associate barcode/tag sequences with designed oligos using the association step of the MPRAsnakeflow pipeline on a subsetted/small dataset.
11:05-11:30	Hands-on: Count Sequencing Analysis We will demonstrate how to count barcodes in both DNA and RNA sequencing on a subsetted/small dataset.
11:30-11:50	Discussion of QC metrics We will discuss the QC parameters and plots returned by MPRAsnakeflow for the full data set, emphasizing the importance of accurate quantification in MPRA experiments.
11:50-12:30	Hands-on: Data Analysis Steps (Regions and Variants) We will walk through potential analytical steps for utilizing count tables and perform statistical analysis using BCalm to identify regions with high activity compared to other sequences in the assay. Additionally, we will highlight methods for identifying variants with significant differences between reference and alternative sequences within designed oligos.
12:30-13:30	Lunch break

Schedule overview (3)

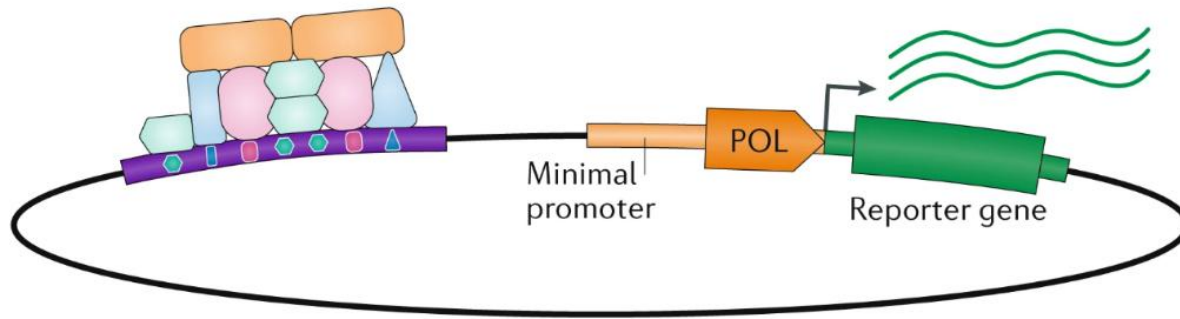
13:30-13:50	Lecture: Modeling of regulatory activity with sequence models We will provide a brief introduction and review of modeling efforts for sequence activity using gapped kmers, convolutional neural networks and language model approaches.
13:50-14:30	Hands-on: Training a sequence based model We will walk through the steps of training a model based on the DeepSTARR CNN architecture.
14:30-14:50	Discussion of pre-training and fine-tuning approaches We discuss the opportunity of pre-training convolutional layers on open chromatin data from multiple cell-types and fine-tuning these convolutions on MPRA data vs. using a language model like NT to train MPRA activity models.
14:50-15:30	Hands-on: Interpreting models with in-silico mutagenesis Supported by tools like TF-MoDISco for motif discovery, participants will investigate important transcription factor binding motifs using models trained on sequence data that predict activity. We will assess whether these motifs exert activating or repressing effects by comparing the activity of sequences with and without these identified motifs in the cell-type of interest.
15:30-16:00	Coffee break

Schedule overview (4)

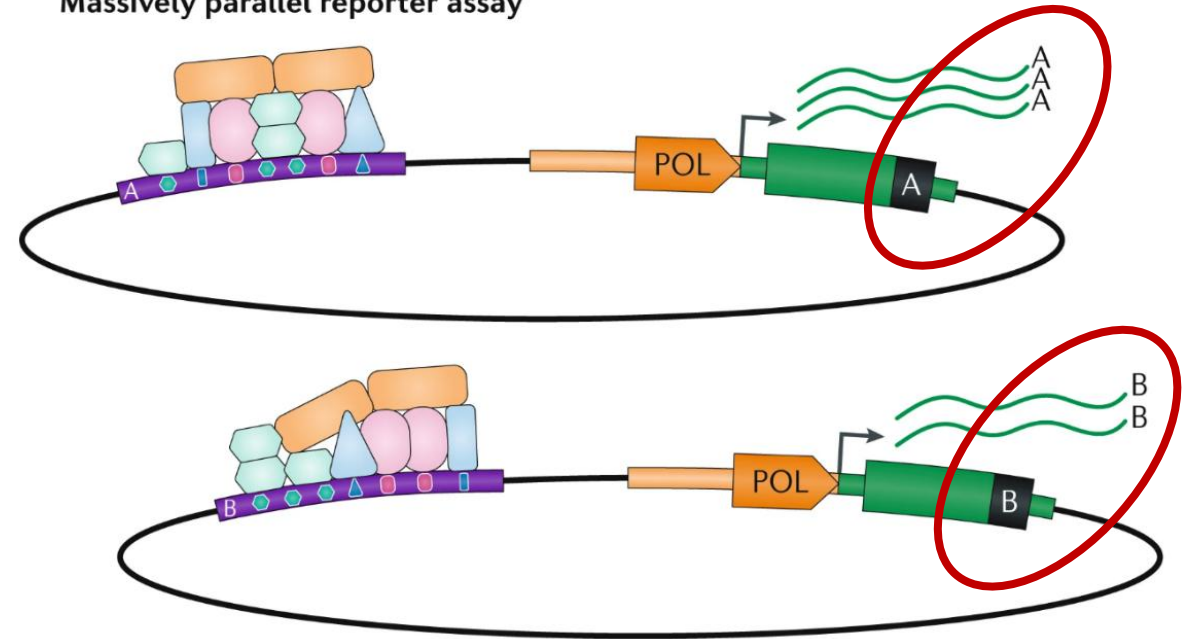
16:00-16:40	Hands-on: Linking motifs to variant effects We will identify transcription factor binding sites (TFBS) overlapping significant variants using motifs derived from model interpretation or available position weight matrices (PWMs) to support the interpretation of variant effects.
16:40-16:50	Discussion of data limitations We discuss the limitations of variant effects in various sequence contexts. We consider the effects of studying only few or related cell-types and the sensitivity of the available modeling approaches for cell-type effects.
16:50-17:20	Participant questions and feedback
17:20-17:30	Concluding remarks by the speakers

Massively parallel reporter assays (MPRAs)

Episomal reporter vector



Massively parallel reporter assay



Also known from other types of reporter gene assays, like Luciferase assays (e.g., promoters, microRNAs)

Gasparini et al, *Nat Reviews Genetics* 2020

Multiplex Assays of Variant Effects (MAVEs) are a valuable resource

Allow study of large sets of sequences/variants and allow relevant biological insight, but:

- Are difficult to perform at scale and require excellent wet lab skills
- Need preparation and planning
- Are still just reporter assays (with biases and artifacts) – e.g., sequences are tested outside of their native genomic context and the read-out is artificial in nature
- MPRA activities are difficult to translate into phenotypic or organismal effects

Provide needed data sets for validation and development of computational predictions

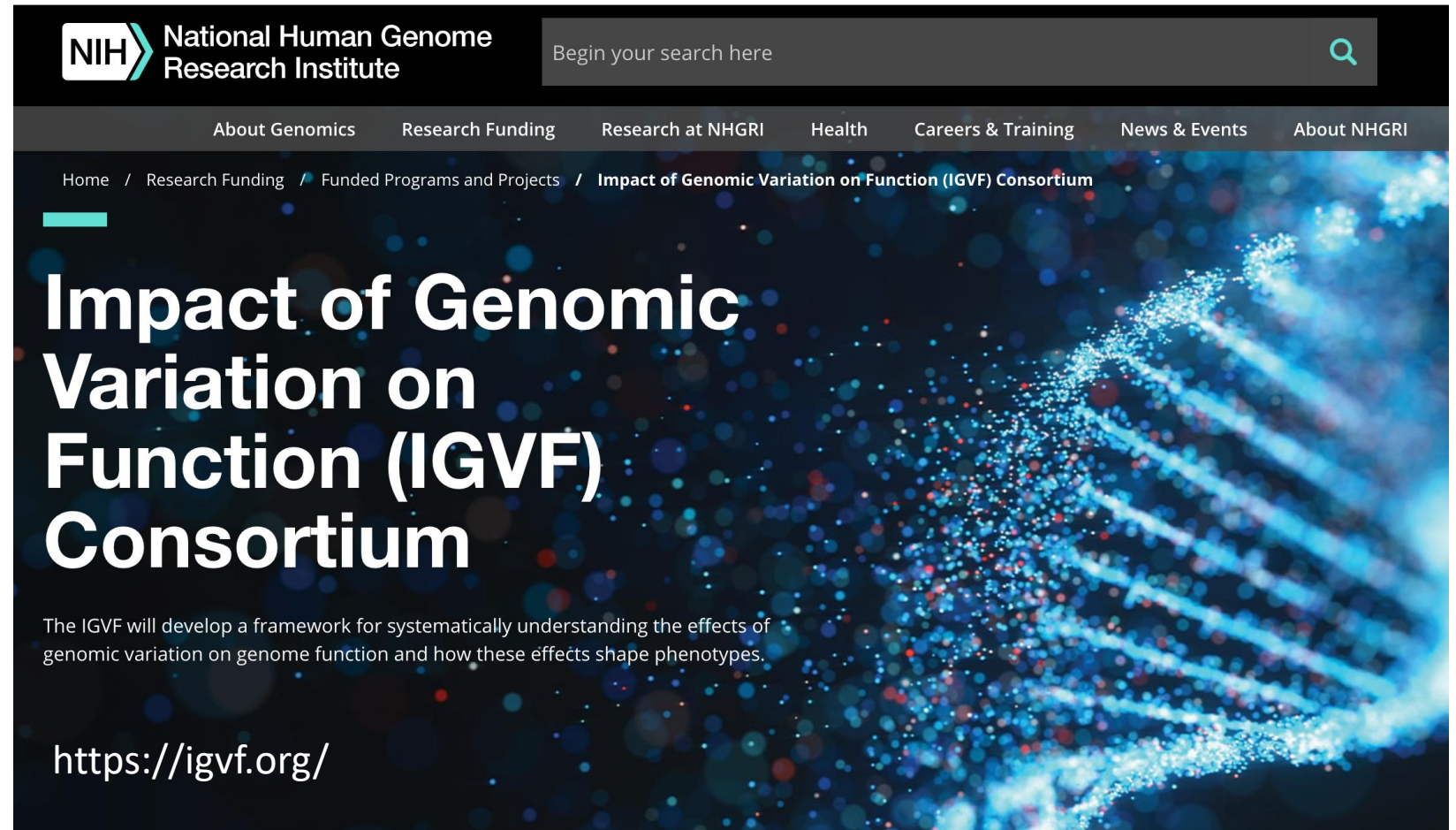
- Available annotations and models are insufficient for variant scoring
- Sequence-based models (CNN/gapped-kmer) limited, but most promising for element activity

Impact of Genomic Variation on Function (IGVF) consortium

Initial funding period 2021-2026, 25 awards (\$185 million) across 30 U.S. research sites plus affiliated (international) groups

Call for renewal expected later this year

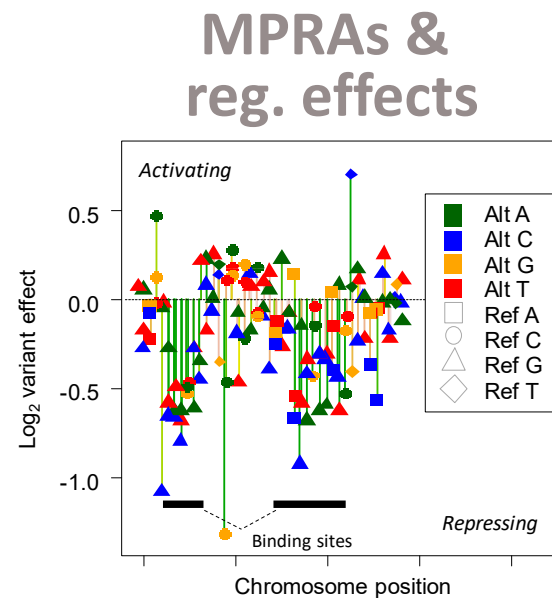
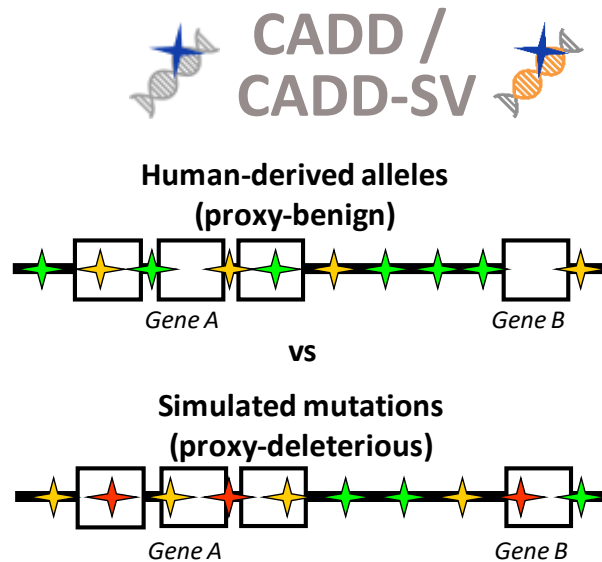
<https://www.genome.gov/Funded-Programs-Projects/Impact-of-Genomic-Variation-on-Function-Consortium>



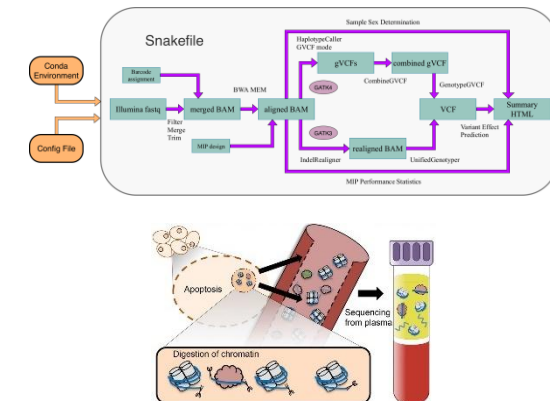
The screenshot shows the NIH National Human Genome Research Institute website. The header includes the NIH logo and the text "National Human Genome Research Institute". A search bar is present with the text "Begin your search here". The navigation menu includes links for "About Genomics", "Research Funding", "Research at NHGRI", "Health", "Careers & Training", "News & Events", and "About NHGRI". The breadcrumb trail reads: "Home / Research Funding / Funded Programs and Projects / Impact of Genomic Variation on Function (IGVF) Consortium". The main heading is "Impact of Genomic Variation on Function (IGVF) Consortium". Below the heading, a paragraph states: "The IGVF will develop a framework for systematically understanding the effects of genomic variation on genome function and how these effects shape phenotypes." At the bottom, the URL <https://igvf.org/> is displayed. The background of the page features a stylized blue and white DNA double helix structure.

Kircher Lab research questions

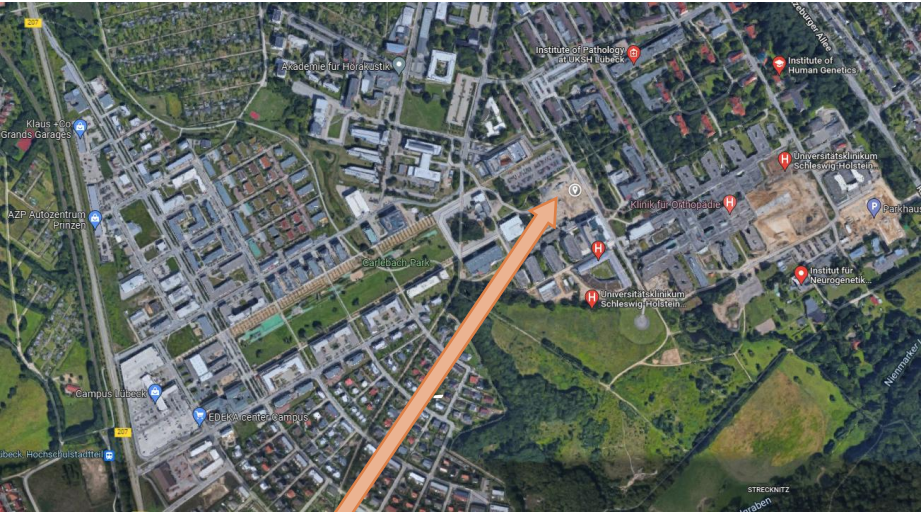
- Genome-wide predictors of variant effects
- Functional read-outs of genomic sequence and its organization
- Supporting the development of molecular assays



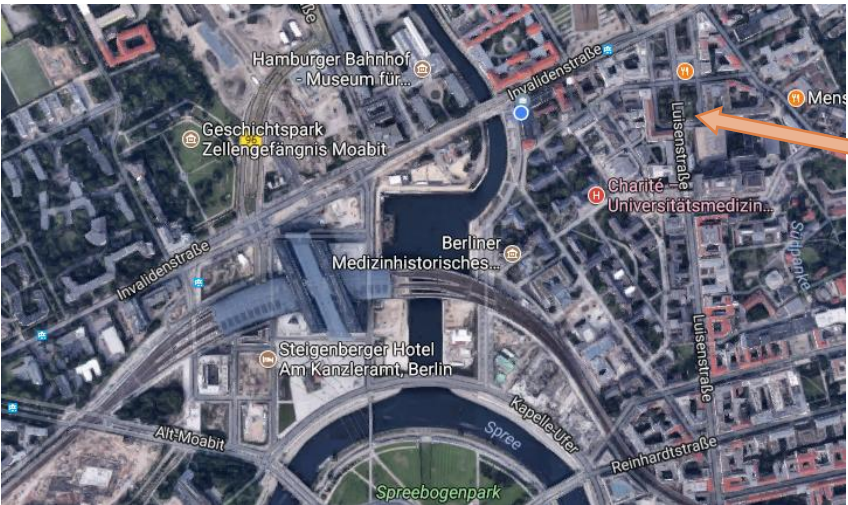
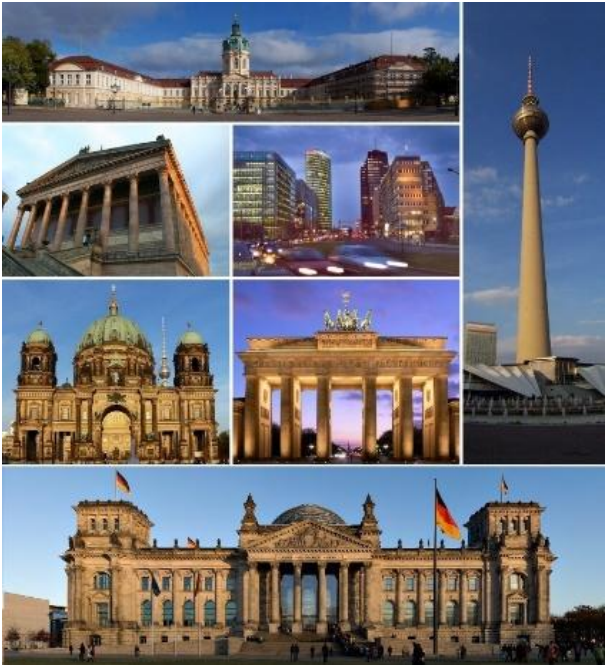
MIPs, cfDNA & others



Kircher Lab in Lübeck & Berlin



BMF Building
UKSH Campus (Lübeck)



RHC at
Charité Campus Mitte
(Berlin)



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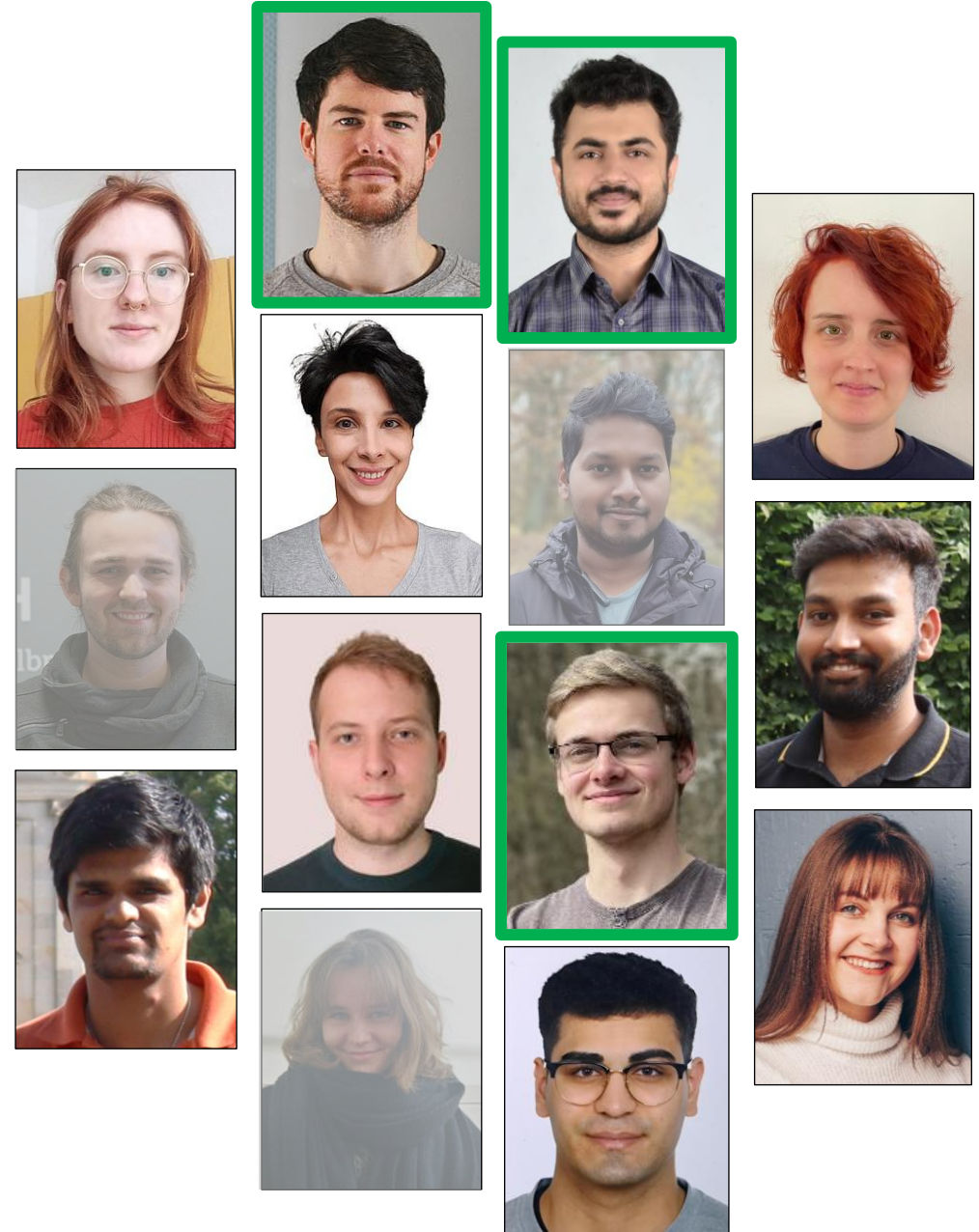


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Schleswig-Holstein



Our Research Team

- PostDocs/Staff:
 - Max Schubach
 - Nikola de Lange
 - Arjun Devadas
- PhD students:
 - Sebastian Röner
 - Lusine Nazaretyan
 - Pyaree Mohan Dash
 - Pia Keukeleire
 - Orazio Catona
 - Kilian Salomon
 - Sudharsan Agas Chandra Prakash
 - Ali Hassan
 - Mustafa Helal
- HiWi / Thesis / Project / Interns students:
 - Kristin Köhler
 - Angelina Göbel-Knapp



Why this tutorial?

What it is supposed to be:

- A platform to meet and exchange
- Examples of how the presenters have done things or are approaching their analyses
- An overview and starting point for discussion
- An opportunity to link to the IGVF consortium

What it is not:

- Comprehensive and authoritative
- A manual of how to run MPRA experimentally
- A step-by-step outline of how to design or analyze MPRA data
- A course on sequence models
- Polished or final

What you should do

- Ask questions!
- Be respectful and eager to learn from others
- Share your thoughts and experiences openly
- Talk to everyone or as many as you can
- Socialize



Questions?



Tutorial materials access and setup of the Google CoLab



ISMB-ECCB 2025: Tutorial IP2 (MPRAs and IGVF Resource)

Max Schubach
Computational Genome Biology @ BIH

July 20, 2025



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All tutorial Materials are linked from here:



- **Access and setup:** Participants will need to access and set up the Google Colab environment.
- **Hands-on exercises:** It is used for hands-on exercises, particularly for **data processing**.
- **Personal laptops:** Participants will engage with these exercises on their **personal laptops**.
- **Resource backup:** A **pre-trained model** is available as a backup for exercises due to potential resource limitations.
- **Setup support:** Help will be provided for participants who encounter difficulties with setup.

https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

Description, links and Colab notebooks for each Part.

Slides of all sessions

Direct links to Colab notebooks

01_introduction	Update README.md	last month
02_MPRAs	Update README.md	last month
03_MPRAsnakeflow	typo fix	2 months ago
04_MPRA_data_analysis	Added outlier analysis, more explanatory text	last month
05_sequence_models	Refactor code structure for improved readability and maintai...	2 months ago
06_variant_effects_and_motifs	06 motif and variant effect: fixed typo in cell	last month
slides	PDF conversion of slides, initial commit	2 weeks ago
README.md	Update README.md	4 days ago

README



ISMB-2025_IGVF-MPRA-Tutorial

This is the main resource repository for the ISMB 2025 Tutorial IP2: Massively parallel reporter assay and regulatory genomics and as part of the IGVF data resource.

PDF conversions of the [preliminary slides are already available](#), but might change after the event and feedback from participants.

We will work with Jupyter notebooks on Google Colab. If you would like to familiarize yourself with it, check out: <https://colab.research.google.com/notebook>

If you want to make sure that you will be able to run these notebooks with your Google account, check out the first [tutorial](#).

Material

https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

01 Introduction





IGVF and its data access

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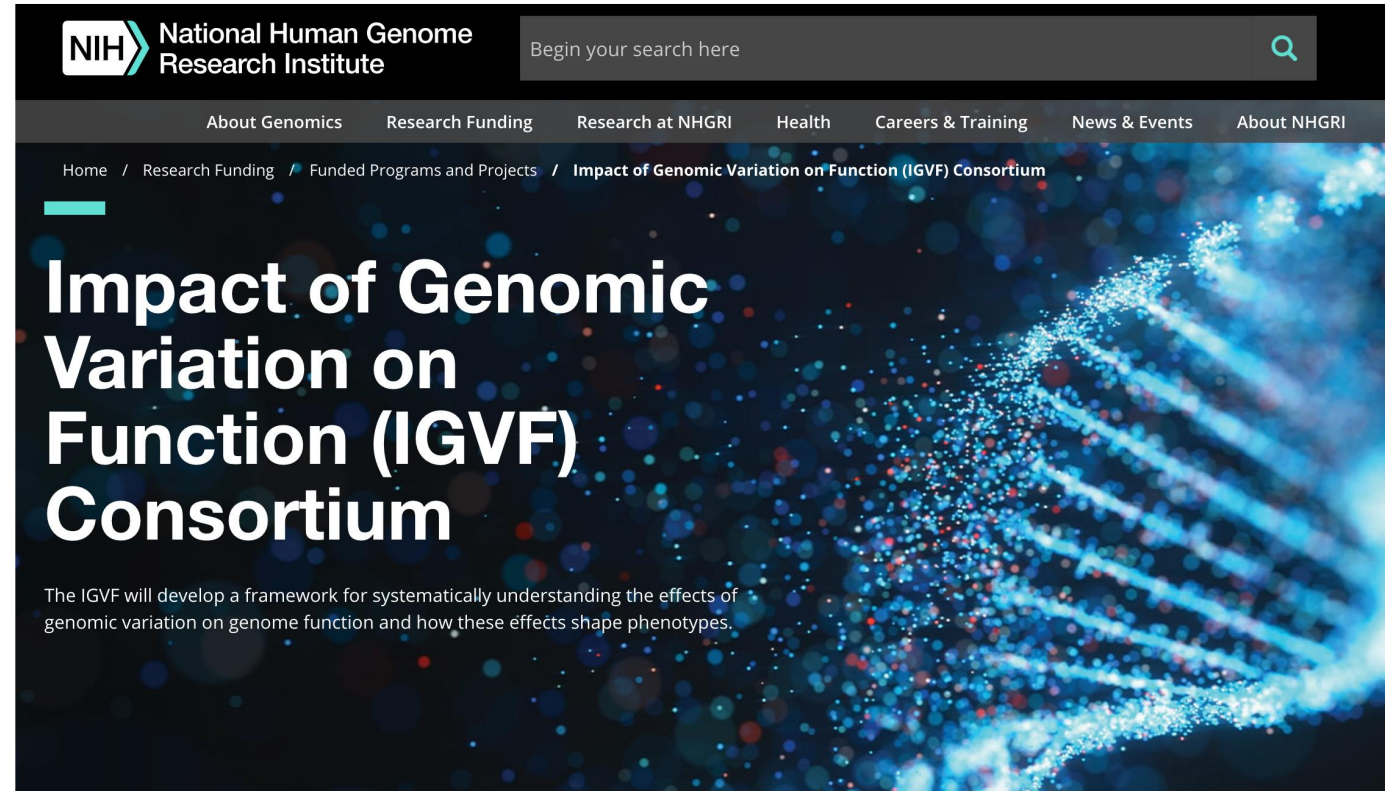
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Impact of Genomic Variation on Function (IGVF) consortium

<https://igvf.org/>

- Initial funding period 2021-2026
- 25 awards (\$185 million) across 30 U.S. research sites plus affiliated (international) groups
- Consortium organized in:
 - Functional Characterization Centers
 - Mapping Centers
 - Regulatory Network Projects
 - Predictive Modeling Projects
 - Data and Administrative Coordinating Centers (DACC)



<https://www.genome.gov/Funded-Programs-Projects/Impact-of-Genomic-Variation-on-Function-Consortium>

Planning year meeting in Washington DC area (2021)



Awardees

Awardee	Institution	Title	Award Number
Characterization Awards			
Nadav Ahituv Martin Kircher Jay Shendure	UC San Francisco Charite Universitatsmedizin Berlin University of Washington	Massively parallel characterization of variants and elements impacting transcriptional regulation in dynamic cellular systems	HG011966
Douglas Fowler Lea Starita	University of Washington	The Center for Actionable Variant Analysis; measuring variant function at scale	HG011969
Jesse Engreitz Thomas Quertermous	Stanford University	Stanford Center for Connecting DNA Variants to Function and Phenotype	HG011972
Marc Vidal	Dana-Farber Cancer Institute	Molecular phenotyping of ~100,000 coding variants across Mendelian disease genes	HG011989
Gary Hon William Kraus Nikhil Munshi	University of Texas Southwestern Medical Center	Multiscale functional characterization of genomic variation in human developmental disorders	HG011996
Michael Love Karen Mohlke Hyejung Won	University of North Carolina at Chapel Hill	Systematic in vivo characterization of disease-associated regulatory variants	HG012003
Daniel Bauer Guillaume Lettre Luca Pinello Richard Sherwood	Children's Hospital Boston Montreal Heart Institute Massachusetts General Hospital Brigham and Women's Hospital	Comprehensive characterization of variants underlying heart and blood diseases with CRISPR base editing	HG012010
Gregory Crawford Charles Gersbach Tim Reddy	Duke University	High-throughput functional annotation of gene regulatory elements and variants critical to complex cellular phenotypes	HG012053



Jay Shendure
(UW)



Nadav Ahituv
(UCSF)



Mike Love
(UNC)

Mapping Awards			
Bradley Bernstein Jason Buenrostro	Broad Institute, Massachusetts General Hospital Broad Institute, Harvard University	A foundational resource of functional elements, TF footprints and gene regulatory interactions	HG011986
Ansuman Satpathy	Stanford University	Single-cell Mapping Center for Human Regulatory Elements and Gene Activity	HG012076
Seyed Mortazavi Barbara Wold	UC Irvine California Institute of Technology	Center for Mouse Genomic Variation at Single Cell Resolution	HG012077
Predictive Modeling Awards			
Alan Boyle	University of Michigan	Predicting the impact of genomic variation on cellular states	HG011952
Andrew S. Allen Shayan Mukherjee Charles D. Page Jr.	Duke University	Design, prediction, and prioritization of systematic perturbations of the human genome	HG011967
Alkes Price Soumya Raychaudhuri Shamil Sunyaev	Harvard School of Public Health Brigham and Women's Hospital Brigham and Women's Hospital	Predicting the impact of genetic variants, genes and pathways on human disease	HG012009
Predrag Radivojac	Northeastern University	Supporting IGVF by modeling genetics, function, and phenotype with machine learning	HG012022
Mark Craven	University of Wisconsin	Linking variants to multi-scale phenotypes via a synthesis of subnetwork inference and deep learning	HG012039
Manuel Garber Xihong Lin Zhiping Weng	University of Massachusetts Medical School Harvard School of Public Health University of Massachusetts Medical School	Predictive modeling of the functional and phenotypic impacts of genetic variants	HG012064
Anshul Kundaje	Stanford University	Predicting context-specific molecular and phenotypic effects of genetic variation through the lens of the cis-regulatory code	HG012069

Network Awards			
Nidhi Sahni Harinder Singh	The University of Texas MD Anderson Cancer Center University of Pittsburgh	Linking genome variation to transcriptional network dynamics in human B cells	HG012041
Sreeram Kannan Hongjun Song Hao Wu	University of Pennsylvania	Defining causal roles of genomic variants on gene regulatory networks with spatiotemporally-resolved single-cell multiomics	HG012047
Michael Beer Anna-Katerina Hadjantonakis Danwei Huangfu	Johns Hopkins University School of Medicine Sloan Kettering Institute for Cancer Research Sloan Kettering Institute for Cancer Research	Genomic control of gene regulatory networks governing early human lineage decisions	HG012051
Hannah Carter Kyle Gaulton Bing Ren Maike Sander	UC San Diego	The impact of genomic variation on environment-induced changes in pancreatic beta-cell states	HG012059
Chongyuan Luo Kathrin Plath Noah Zaitlen	UC Los Angeles	Leveraging genetic variation to dissect gene regulatory networks of reprogramming to pluripotency	HG012079
Christina Leslie Alexander Rudensky	Sloan Kettering Institute for Cancer Research	Deciphering the genomics of gene network regulation of T cell and fibroblast states in autoimmune inflammation	HG012103
Data and Administrative Coordinating Center Awards			
J. Michael Cherry Mark Gerstein	Stanford University Yale University	A Data and Administrative Coordinating Center for the Impact of Genomic Variation on Function Consortium	HG012012
Ting Wang Feng Yue	Washington University, Saint Louis Northwestern University	WashU-Northwestern Genomic Variation and Function Data and Administrative Coordinating Center	HG012070

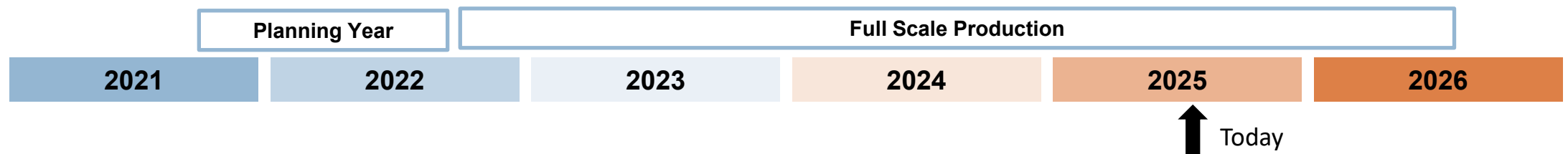
Network Awards				
Nidhi Sahni Harinder Singh	The University of Texas MD Anderson Cancer Center University of Pittsburgh	Linking genome variation to transcriptional network dynamics in human B cells		HG012041
Sreeram Kannan Hongjun Song Hao Wu	University of Pennsylvania	Defining causal roles of genomic variants on gene regulatory networks with spatiotemporally-resolved single-cell multiomics		HG012047
Michael Beer Anna-Katerina Hadjantonakis Danwei Huangfu	Johns Hopkins University School of Medicine Sloan Kettering Institute for Cancer Research Sloan Kettering Institute for Cancer Research	Genomic control of gene regulatory networks governing early human lineage decisions		HG012051
Hannah Carter Kyle Gaulton Bing Ren Maike Sander	UC San Diego	Dissecting gene regulatory networks governing cell fate transitions on environment-dependent cell states		HG012059
Chongyuan Luo Kathrin Plath Noah Zaitlen	UC Los Angeles	Dissecting gene regulatory networks of reprogramming to pluripotency		HG012079
Christina Leslie Alexander Rudensky	Sloan Kettering Institute for Cancer Research	Deciphering the genomics of gene network regulation of T cell and fibroblast states in autoimmune inflammation		HG012103
Data and Administrative Coordinating Center Awards				
J. Michael Cherry Mark Gerstein	Stanford University Yale University	A Data and Administrative Coordinating Center for the Impact of Genomic Variation on Function Consortium		HG012012
Ting Wang Feng Yue	Washington University, Saint Louis Northwestern University	WashU-Northwestern Genomic Variation and Function Data and Administrative Coordinating Center		HG012070

+ about 30 affiliated groups (incl. non-US)

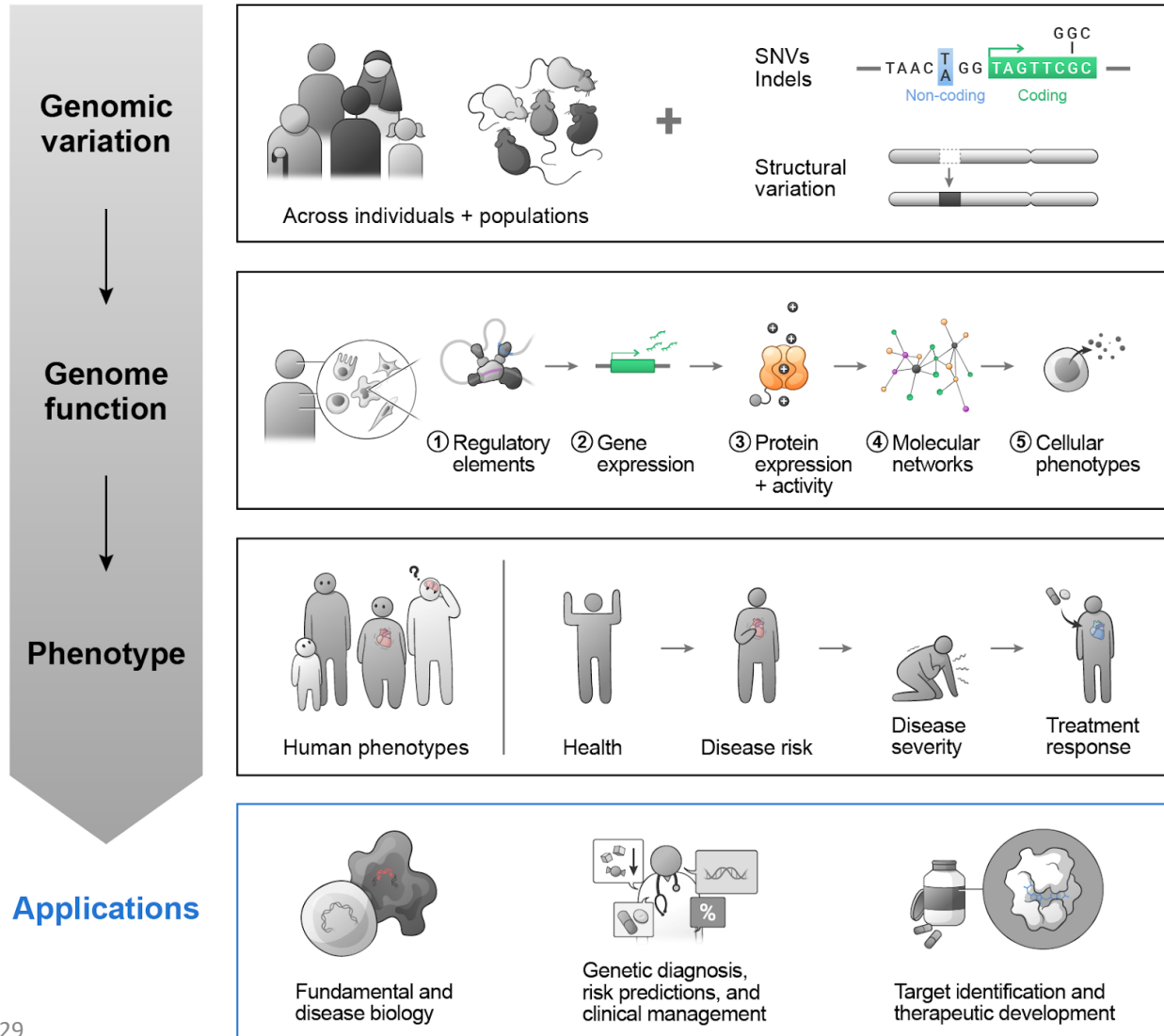
Impact of Genomic Variation on Function



- NIH/NHGRI consortium
 - Goal: Transforming our understanding of how variation impacts function and leads to phenotypes in health and disease
 - Using systematic perturbation to assess genome function
 - Development of predictive models of genome function
 - Identification of where and when genes and regulatory elements function at high resolution
 - Network-level understanding of genome function
 - ***Generation of a catalog of elements, variants and phenotypes; sharing of data, predictions, models, and tools***
 - ***Enabling others to apply these approaches***



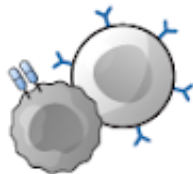
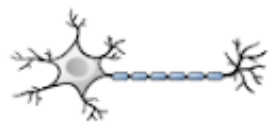
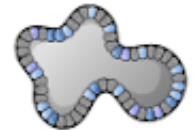
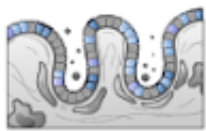
Questions



Focus areas	Example
Coding variation	Which missense VUS are likely pathogenic?
Non-coding variation	Which variants affect the expression of which genes?
Molecular networks	What are the direct target genes of transcription factors?
Disease	Which variants, cell-types, genes and pathways influence disease risk?

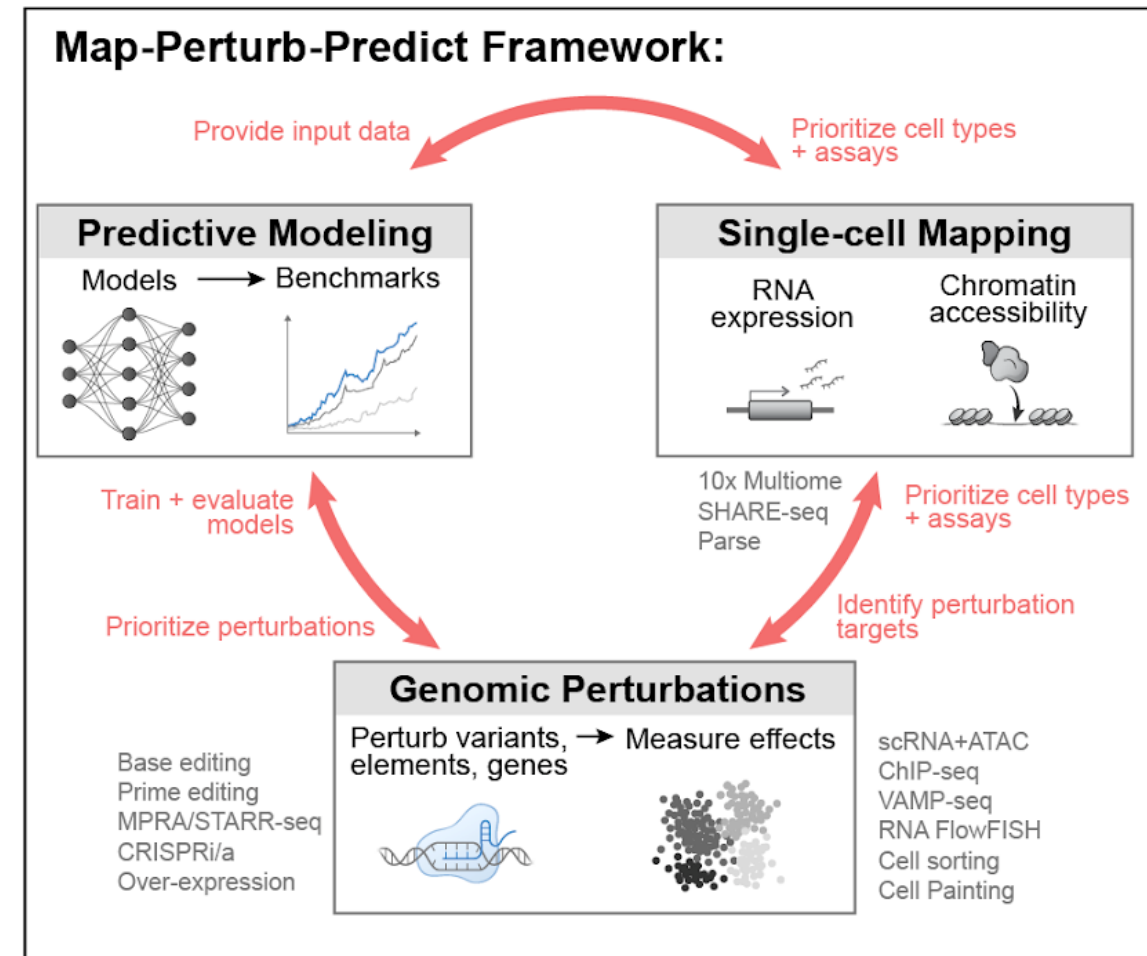
<https://doi.org/10.1038/s41586-024-07510-0>

Biological Systems & diseases

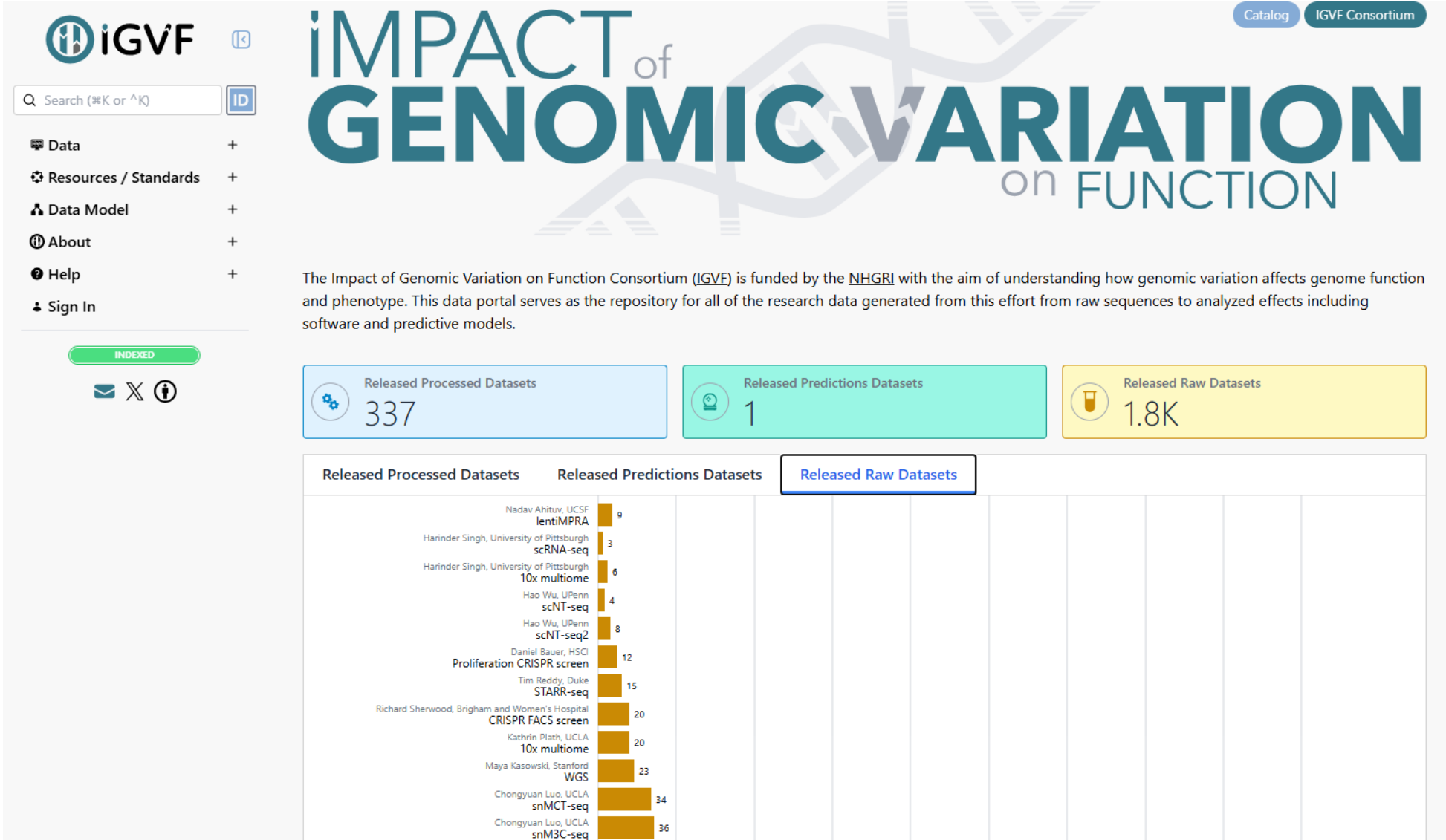
Systems + Diseases		Samples	
	Coronary artery disease Type 2 diabetes Lipid traits Blood pressure Congenital heart defects		K562, GM12878 HUDEP-2, HAP1 ENCODE cell lines B cells Smooth muscle cells
Cardiometabolic		Cell lines + primary cells	
	Systemic lupus erythramatosus Rheumatoid arthritis Red blood cell traits		Cardiomyocytes Endothelial cells Fibroblast to iPSC Macrophages Neurons Pancreatic cells Trophoblasts
Immune + hematopoietic		hPSC differentiation + reprogramming	
	Parkinson's disease Alzheimer's disease Neurodevelopmental delay		...
Neuro		Organoids	Cardioids Cerebral organoids Gastruloids
	Williams Syndrome DiGeorge Syndrome Hereditary cancers		Aorta Blood Bone marrow Brain Heart Liver
Developmental + syndromic		Tissues	...

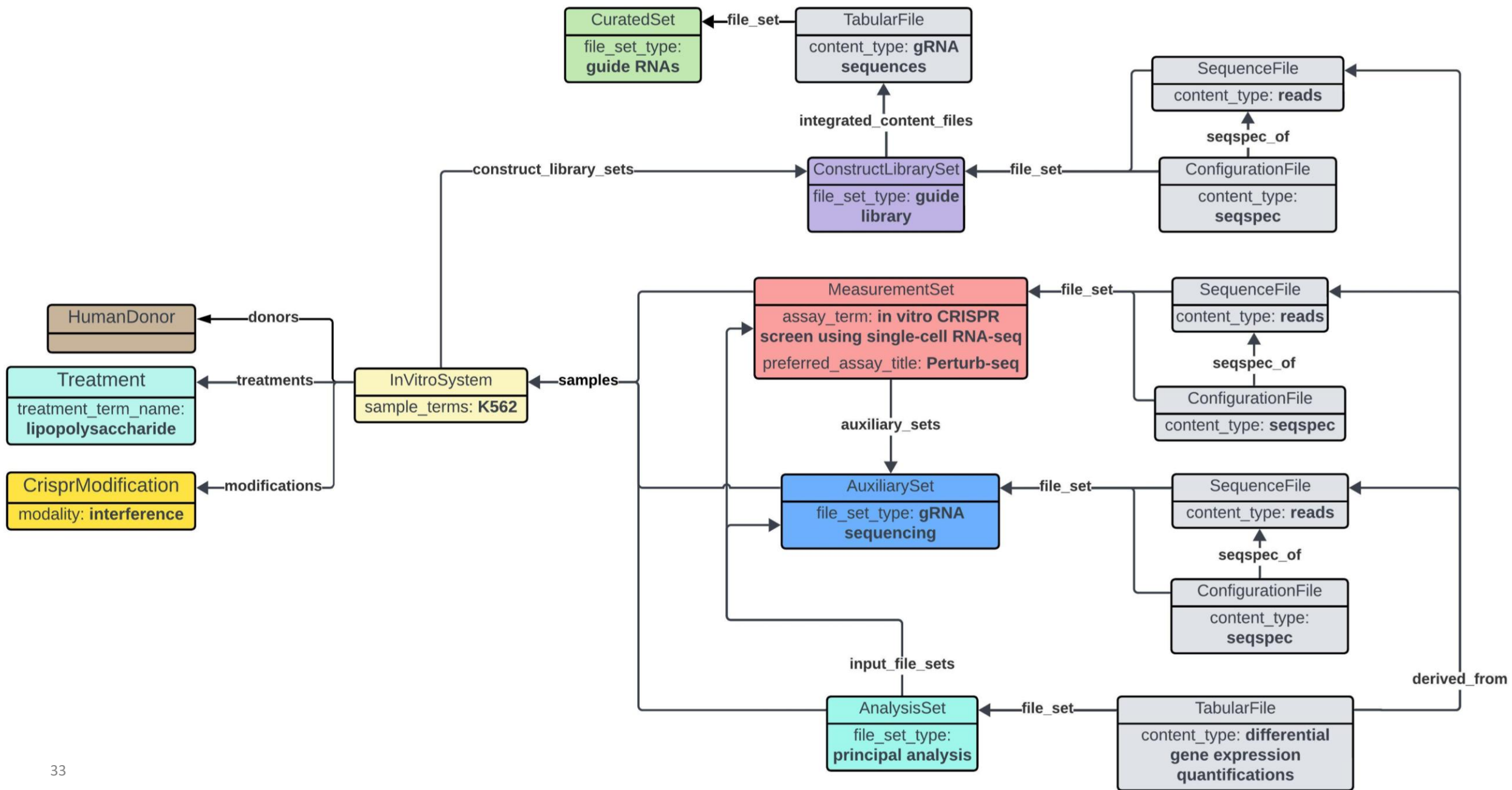
Data and Analyses

- Mapping: transcriptomes, chromatin accessibility
- Perturbations (genetic and epigenetic): MAVEs (e.g. MPRA, DMS), CRISPR
- Predictive modeling: unseen variants
- Network/pathway analysis: Gene Regulatory Networks (GRNs), Protein Protein Interactions (PPIs)
- Catalog of Variant/Element/Phenotype connections



IGVF data portal: <https://data.igvf.org/>







Q Search (%K or ^K)

ID

Data

Raw Data Sets

Processed Data Sets

Prediction Data Sets

Files

Resources / Standards

Data Model

Schemas

Data Model Examples

About

Help

Sign In

HOME / MEASUREMENT SETS / IGVFDS5476WHHS

IGVFDS5476WHHS

RELEASED

</>

File Set Type

experimental data

Summary

lentiMPRA integrating a reporter library targeting candidate cis-regulatory elements genome-wide

Description

HepG2 RNA count data of replicate 3

Assay Term

[massively parallel reporter assay](#)

Sample Summaries

Homo sapiens (male) HepG2 pooled cell specimen transduced (lentivirus) with a reporter library

Protocols

<https://www.protocols.io/view/massive-parallel-reporter-assay-mpa-kxygxpmkw18j/v1>

Submitter Comment

rep3

Attribution

[Nadav Ahituv, UCSF \(HG011966 — Jay Shendure\)](#)

Sequencing Results (Illumina)

9 items

Accession	File Format	Content Type	Illumina Read Type	Sequencing Run	Read Names	Associated seqspec Files	Sequence Specification Document	Sequencing Platform	Flowcell ID	Lane
IGVFFI0130HTTZ  	fastq	reads	R1	3	Barcode forward	IGVFFI4045BFNH  		Illumina NextSeq 500		
IGVFFI1164KXKU  	fastq	reads	R3	3	Barcode reverse	IGVFFI4045BFNH  		Illumina NextSeq 500		
IGVFFI3537XINM  	fastq	reads	R3	1	Barcode reverse	IGVFFI4045BFNH  		Illumina NextSeq 500		

INDEXED



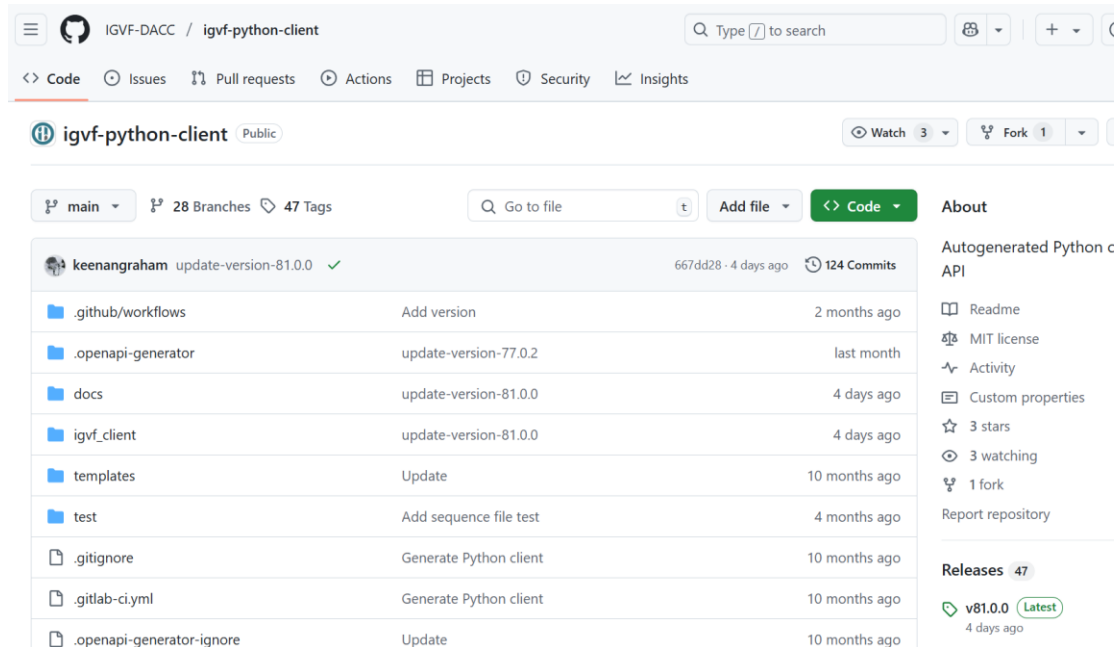
34

You can also download from the portal using IGVF tools or an API

https://github.com/IGVF-DACC/igvf_utils

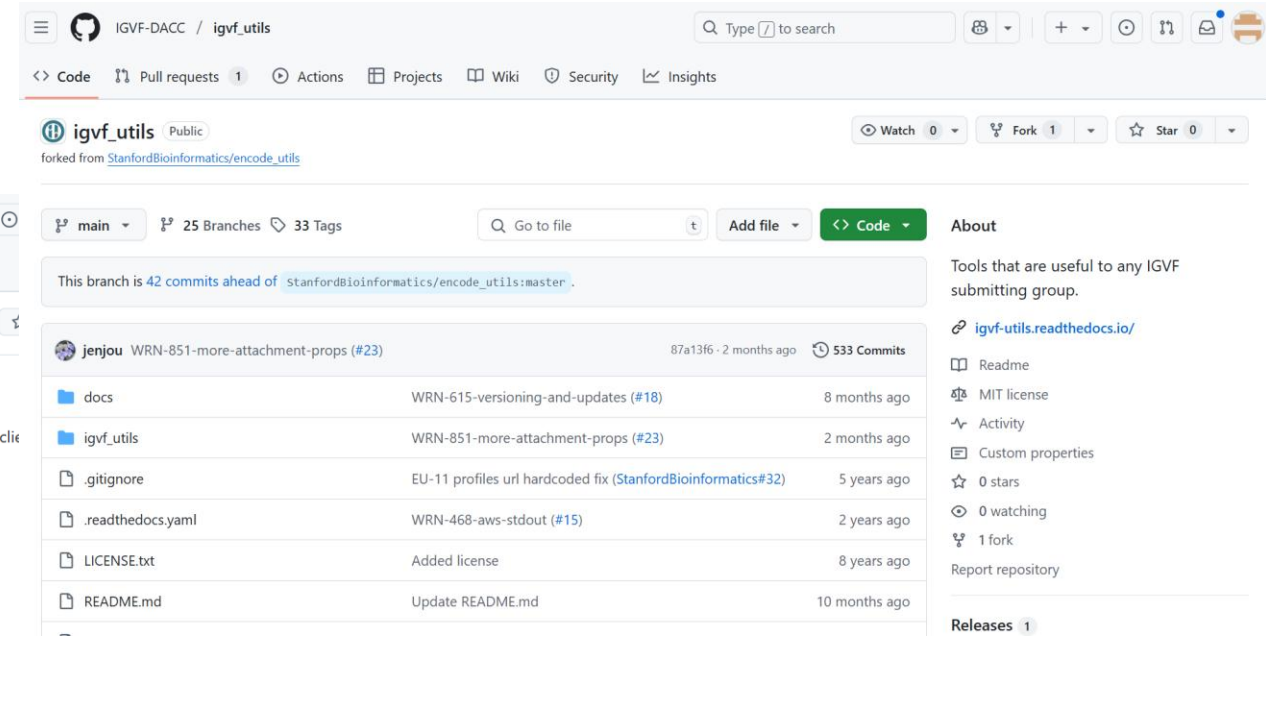
Getting started: https://github.com/IGVF-DACC/igvf_utils/wiki

<https://github.com/IGVF-DACC/igvf-python-client>



The screenshot shows the GitHub repository page for `igvf-python-client`. The repository is public and has 28 branches and 47 tags. The main branch is selected. The repository is described as "Autogenerated Python client API". The repository has 124 commits, with the latest commit being `update-version-81.0.0` by `keenangraham` 4 days ago. The repository is licensed under MIT. The repository has 3 stars and 1 fork. The repository is being watched by 3 people. The repository is being forked by 1 person. The repository is being starred by 0 people. The repository is being watched by 0 people. The repository is being forked by 1 person. The repository is being starred by 0 people.

File	Commit	Time
<code>.github/workflows</code>	Add version	2 months ago
<code>.openapi-generator</code>	update-version-77.0.2	last month
<code>docs</code>	update-version-81.0.0	4 days ago
<code>igvf_client</code>	update-version-81.0.0	4 days ago
<code>templates</code>	Update	10 months ago
<code>test</code>	Add sequence file test	4 months ago
<code>.gitignore</code>	Generate Python client	10 months ago
<code>.gitlab-ci.yml</code>	Generate Python client	10 months ago
<code>.openapi-generator-ignore</code>	Update	10 months ago



The screenshot shows the GitHub repository page for `igvf_utils`. The repository is public and is a fork of `StanfordBioinformatics/encode_utils`. The repository has 25 branches and 33 tags. The main branch is selected. The repository is described as "Tools that are useful to any IGVF submitting group." The repository has 533 commits, with the latest commit being `WRN-851-more-attachment-props (#23)` by `jenjou` 2 months ago. The repository is licensed under MIT. The repository has 0 stars and 1 fork. The repository is being watched by 0 people. The repository is being forked by 1 person. The repository is being starred by 0 people. The repository is being watched by 0 people. The repository is being forked by 1 person. The repository is being starred by 0 people.

File	Commit	Time
<code>docs</code>	WRN-615-versioning-and-updates (#18)	8 months ago
<code>igvf_utils</code>	WRN-851-more-attachment-props (#23)	2 months ago
<code>.gitignore</code>	EU-11 profiles url hardcoded fix (StanfordBioinformatics#32)	5 years ago
<code>.readthedocs.yml</code>	WRN-468-aws-stdout (#15)	2 years ago
<code>LICENSE.txt</code>	Added license	8 years ago
<code>README.md</code>	Update README.md	10 months ago

variant / element quantifications

Search (%K or ^K)

ID

Data

Raw Data Sets

Processed Data Sets

Prediction Data Sets

Files

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IGVFDS5394QLIN

RELEASED

File Set Type

principal analysis

Summary

lentiMPRA integrating a reporter library targeting histone modifications, transcription start sites, phenotype-associated variants genome-wide associated with autism spectrum disorder

Publications

[Characterization of De Novo Promoter Variants in Autism Spectrum Disorder with Massively Parallel Reporter Assays](#)

Samples

Homo sapiens H1-hESC pooled differentiated cell specimen induced to glutamatergic neuron, at 7 day(s) post change, transfected with a reporter library

Protocols

<https://www.protocols.io/view/massive-parallel-reporter-assay-mpa-kxygxpmkw18j/v1>

Attribution

[Nadav Ahituv, UCSF \(HG011966](#) — [Jay Shendure](#)

Files

1 item

Accession	File Format	Content Type	Summary	Lab	File Size	Workflow	Uniform Pipeline	Upload Status
IGVFFI6886KJDH	tsv	reporter genomic variant effects	hg19 reporter genomic variant effects	Nadav Ahituv, UCSF	121 KB	MPRAnalyze Variant Annotation		VALIDATED

File Association Graph

IGVFDS9668QVZD
3 files
reporter library

IGVFFI4651HMEA
tsv
barcode to element...

File Set

File

Analysis Sets

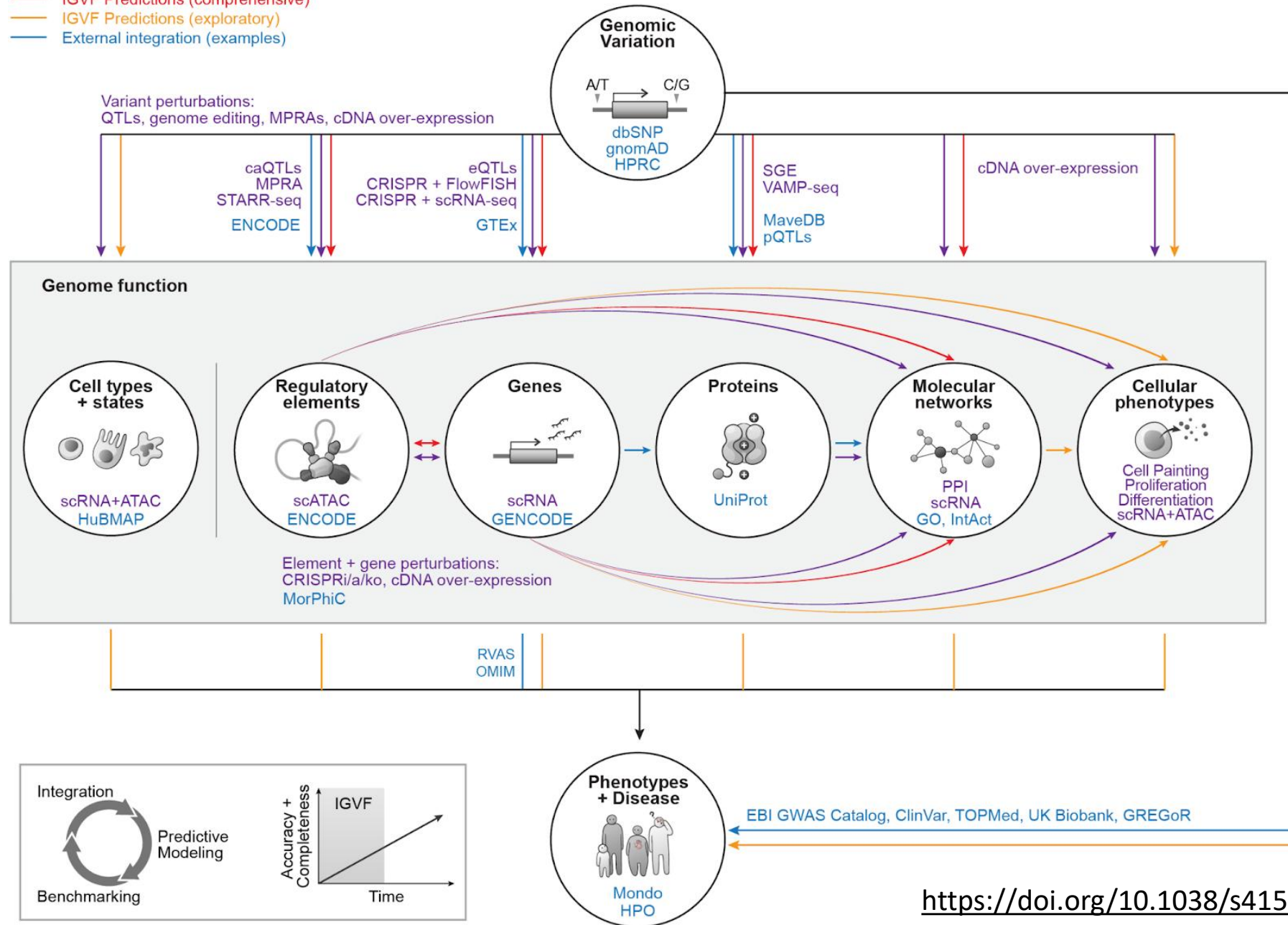
Auxiliary Sets

Curated Sets

Measurement Sets

IGVF Catalog:

- IGVF Experiments (examples)
- IGVF Predictions (comprehensive)
- IGVF Predictions (exploratory)
- External integration (examples)

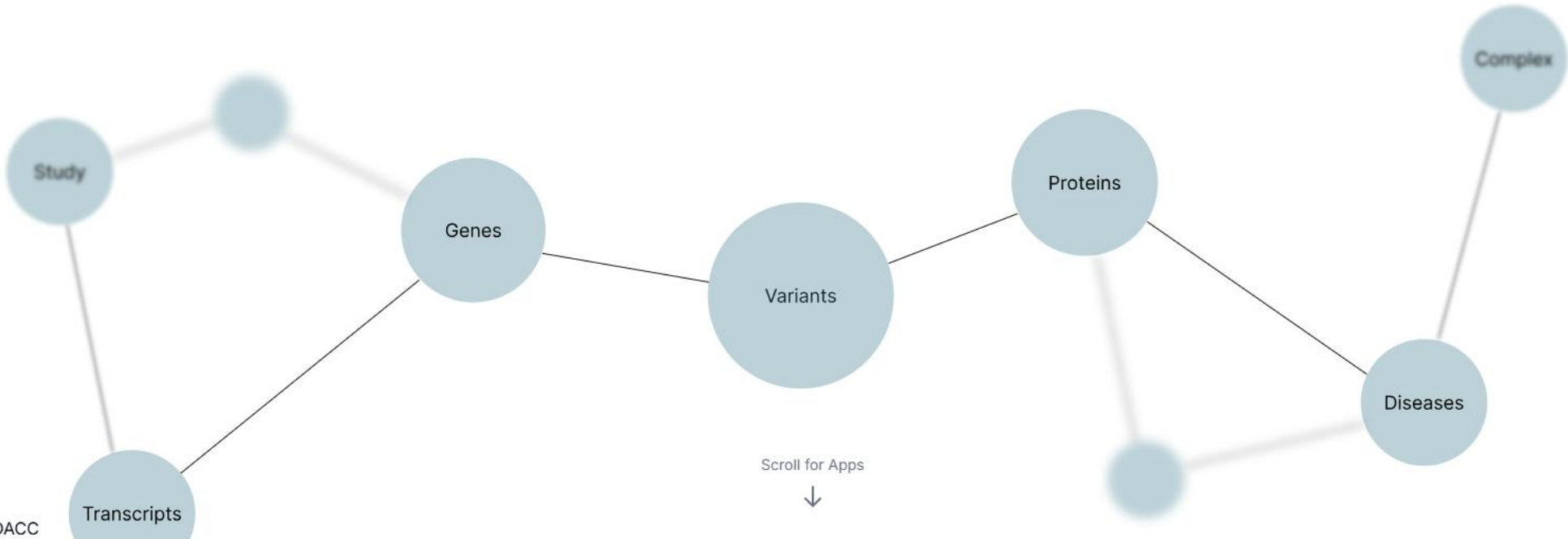


Search by anything.

Q

rs7582141, TP53, 1433B_HUMAN, cancer ...

 rs1250566  ENSG00000187642  rs334  17-46935905-T-C  rs5865



Scroll for Apps
↓

Variant rs1250566 (G-A)Variant rs1250566 (G-T)

☐ X-Ray

Live Graph

SPDI: NC_000010.11:79286695:G:A | HGVS: NC_000010.11:g.79286696G>A | Gnomad AF: 0.23977 | [10-79286695-G-A](#)

Jump to: [Enhancer-Gene Prediction](#) | [Variants in Linkage Disequilibrium](#) | [GWAS Association](#) | [Coding Variant Prediction](#) | [Allelic effect\(s\) on transcription factor binding](#) | [pQTL](#) | [QTL](#) | [Associated Genes and Pr](#)

203 Enhancer-Gene Predictions

Which genes are predicted to be regulated by enhancers overlapping this variant?

Download: JSON CSV

Search in table...

Cell Type	Target Gene	Score	Dataset	Model	Variant-Gene ...
gm19238	PPIF	0.947	files	ENCODE-E2G-DNaseOnly	60773
gm18507	PPIF	0.945	files	ENCODE-E2G-DNaseOnly	60773
gm19240	PPIF	0.937	files	ENCODE-E2G-DNaseOnly	60773
gm19239	PPIF	0.922	files	ENCODE-E2G-DNaseOnly	60773
gm13976	PPIF	0.887	files	ENCODE-E2G-DNaseOnly	60773
gm10248	PPIF	0.880	files	ENCODE-E2G-DNaseOnly	60773
gm10266	PPIF	0.873	files	ENCODE-E2G-DNaseOnly	60773
cd14-positive monocyte	PPIF	0.870	files	ENCODE-E2G-DNaseOnly	60773
naive b cell	PPIF	0.870	files	ENCODE-E2G-DNaseOnly	60773
gm12891	PPIF	0.867	files	ENCODE-E2G-DNaseOnly	60773

37 Allelic effect(s) on transcription factor bindings

Effects of variant on allelic binding in transcription factor (TF) ChIP-seq experiments



Download: JSONCSV

Search in table...

Protein	Gene	Biosample	Motif	Motif Summary	Motif Fold-Change	Ref Score	Alt Score	Ref FDR	Alt FDR	Motif P
ENSP00000227163	SPI1	monocyte-derived macrophages from peripheral blood	SPI1_HUMAN_HOCOMOCov11	No Hit	0.746	1.619	-1.280	0.028	1.000	6
ENSP00000263377	BRD4	primary B-CLL cells	BRD4_HUMAN_HOCOMOCov11	No Hit	0.000	1.811	-1.170	0.014	1.000	
ENSP00000343896	SMARCA4	J-Lat A72	SMCA4_HUMAN_HOCOMOCov11	No Hit	0.000	-2.350	2.077	1.000	0.007	
ENSP00000353112	BRD4	primary B-CLL cells	BRD4_HUMAN_HOCOMOCov11	No Hit	0.000	1.811	-1.170	0.014	1.000	
ENSP00000353504	RUNX1T1	Kasumi-1 (acute myeloblastic leukemia)	MTG8_HUMAN_HOCOMOCov11	No Hit	0.000	-1.978	0.397	1.000	0.023	
ENSP00000354394	STAT1	CD14+ monocytes	STAT1_HUMAN_HOCOMOCov11	No Hit	0.029	0.957	-1.660	0.002	1.000	9
ENSP00000360901	BRD4	primary B-CLL cells	BRD4_HUMAN_HOCOMOCov11	No Hit	0.000	1.811	-1.170	0.014	1.000	
ENSP00000367799	SPI1	monocyte-derived macrophages from peripheral blood	SPI1_HUMAN_HOCOMOCov11	No Hit	0.746	1.619	-1.280	0.028	1.000	6
ENSP00000376136	STAT1	CD14+ monocytes	STAT1_HUMAN_HOCOMOCov11	No Hit	0.029	0.957	-1.660	0.002	1.000	9
ENSP00000376137	STAT1	CD14+ monocytes	STAT1_HUMAN_HOCOMOCov11	No Hit	0.029	0.957	-1.660	0.002	1.000	9

0 pQTLs

Protein quantitative trait loci associated with this variant



Gene PPIF

☐ X-Ray

Live Graph

Jump to: [Enhancer-Gene Prediction](#) | [Pathway](#) | [Gene Structure](#) | [Gene Interactions & Coexpression](#)

23428 Enhancer-Gene Predictions

Which enhancers are predicted to regulate this gene, and in which cell types?

Download: [JSON](#) [CSV](#)

Search in table...

Cell Type	▼ Score	Start	End	Dataset	Model	Distance (Enhancer to Gene)
brain	1.000	79347213	79347713	ENCFF679SQC	ENCODE-E2G-DNaseOnly	0
testis	1.000	79347213	79347713	ENCFF262XSJ	ENCODE-E2G-DNaseOnly	0
gastrocnemius medialis	1.000	79347212	79347713	ENCFF951VUI	ENCODE-E2G-DNaseOnly	0
ag20443 cell	1.000	79347208	79347713	ENCFF647HXH	ENCODE-E2G-DNaseOnly	0
d341med	1.000	79347207	79347713	ENCFF403ZFY	ENCODE-E2G-DNaseOnly	0
t47d	1.000	79347209	79347713	ENCFF712TNB	ENCODE-E2G-DNaseOnly	0
gm13976	1.000	79347206	79347713	ENCFF575YOJ	ENCODE-E2G-DNaseOnly	0
lung	1.000	79347197	79347713	ENCFF272HRW	ENCODE-E2G-DNaseOnly	0
imr-90	1.000	79347202	79347713	ENCFF685OHY	ENCODE-E2G-DNaseOnly	0
left lung	1.000	79347202	79347713	ENCFF852ZMN	ENCODE-E2G-DNaseOnly	0

Explore yourself!

It is a lot more than (just) MPRA data

<https://data.igvf.org/>

<https://catalog.igvf.org/>

There is also an API:

<https://api.catalog.igvf.org/>

& bioconductor R package:

<https://bioconductor.org/packages/devel/bioc/vignettes/rigvf/inst/doc/rigvf.html>

GET

/variants/summary

▼

GET

/variants/proteins

▼

GET

/variants/diseases

▲

Retrieve diseases and genes associated with the query variant from ClinGen.

Example: variant_id = e4b5a3b5c96984f03ed0a79dca6342d3d74cbc642ae1ea589f409c04ccc3044f

spdi = NC_000012.12:102917129:T:C,

hgvs = NC_000012.12:g.102917130T>C,

rsid = rs62514891,

chr = chr12,

position (zero base) = 102917129,

pmid = 2574002.

The limit parameter controls the page size and can not exceed 100.

Pagination is 0-based.

Questions?



Thank You! Questions?

kircherlab.github.io

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